

LUO WENJIE (罗文杰)

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PROFESSIONAL SUMMARY

MSc Civil Engineering candidate at NUS with a foundation in structural analysis, Lean Construction, contract administration, and transport engineering; trained on FIDIC with working knowledge of Singapore standard forms (PSSCOC, SIA Standard Form) and the BCA codes. Bilingual in Mandarin and English, with demonstrated team collaboration and structured problem-solving across multidisciplinary group projects. Seeking a full-time internship (June 2026 – January 2027) and subsequent Graduate Engineer position with China Communications Construction Company (Singapore), contributing to civil and infrastructure project delivery in Singapore and the wider region.

EDUCATION

National University of Singapore (NUS) — Singapore

Aug 2025 – Jan 2027

Master of Science in Civil Engineering

- **Construction & Management:** Lean Construction Practice (CE5806), Construction Project Planning & Site Management, Transport Infrastructure Asset Management (CE5208), Intermodal Transportation (CE5212).
- **Structural & Analytical:** Finite Element Concepts & Applications (CE5010), Advanced Structural Steel Design (CE5509), Design of Composite Steel and Concrete Structures, Traffic Flow & Control (CE5203), Research Project (CE5001) — supervised by Prof. Ong Ghim Ping Raymond and Dr. Geng Guoqing.

Chang'an University (CHD) — Xi'an, China

Sep 2021 – Jun 2025

Bachelor of Engineering in Transportation Engineering — Outstanding Engineer Programme | GPA 3.17 / 4.00

- **Key Coursework:** Finite Element Analysis, Subgrade and Pavement Engineering, Road Engineering Theory, Transportation Engineering Theory, Traffic Survey & Analysis.

CORE SKILLS

- **Construction & Contracts:** Construction Scheduling | Resource Allocation | Delay Analysis | Critical-Path Mitigation | Last Planner System | Work Sampling | Crew Balance Charts | Seven Mudras (Lean) | FIDIC | PSSCOC | SIA Standard Form | BCA Codes | EOT & Variation Orders.
- **Engineering Software:** SAP2000 | AutoCAD | SUMO (microscopic traffic simulation) | Python | MATLAB | LaTeX | MS Office (Word, Excel, PowerPoint).
- **Languages:** Mandarin (native), English (professional working proficiency).

SELECTED ACADEMIC PROJECTS

Multimodal Transport Network Resilience to Floods — MSc Research Project (CE5001)

Aug 2025 – Jan 2027

- Developing a coupled flood and transport-network model under Prof. Ong Ghim Ping Raymond and Dr. Geng Guoqing to analyse the dynamic impact of flood events on transport-network operations, supporting urban infrastructure risk-management decisions.

SUMO Microscopic Traffic Simulation — AYE Westbound Weaving Section

Feb 2026 – Apr 2026

Group Project, NUS CE5203

- Built a calibrated sublane (SL2015) microsimulation model of the Ayer Rajah Expressway Exit 9–11 weaving section in SUMO; generated demand files via Python and tuned speedFactor distributions to match field data.
- Computed four performance metrics (throughput, delay, stops, density) for peak and off-peak periods; produced a React/Recharts dashboard for results visualisation and co-authored Sections 3.3, 3.4, and 4 of the formal project report.

Lean Waste Audit — Timber Formwork Erection

Feb 2026 – Apr 2026

Group Project, NUS CE5806

- Conducted work-sampling and crew-balance analysis of a recorded timber formwork operation; identified push-based overproduction and idle-time waste, recommending a seamless crew-rebalancing solution to eliminate ~190 s of idle time per cycle.

Finite Element Analysis of Plate Structures (SAP2000)

Aug 2025 – Sep 2025

Course Project, NUS

- Modelled an aluminium plate subjected to complex distributed and angled point loads; ran linear FE analysis comparing Constant Strain Triangle and 4-node Quadrilateral elements, with mesh-convergence study to validate displacement and von-Mises stress results.

PRACTICAL EXPERIENCE

Field Surveyor — Chang'an University Course Practice

Dec 2021 – Jan 2022

- Conducted plane and elevation control surveys at designated sites using total stations and levelling instruments.
- Performed office-based calculations to verify survey data and mitigate measurement errors.